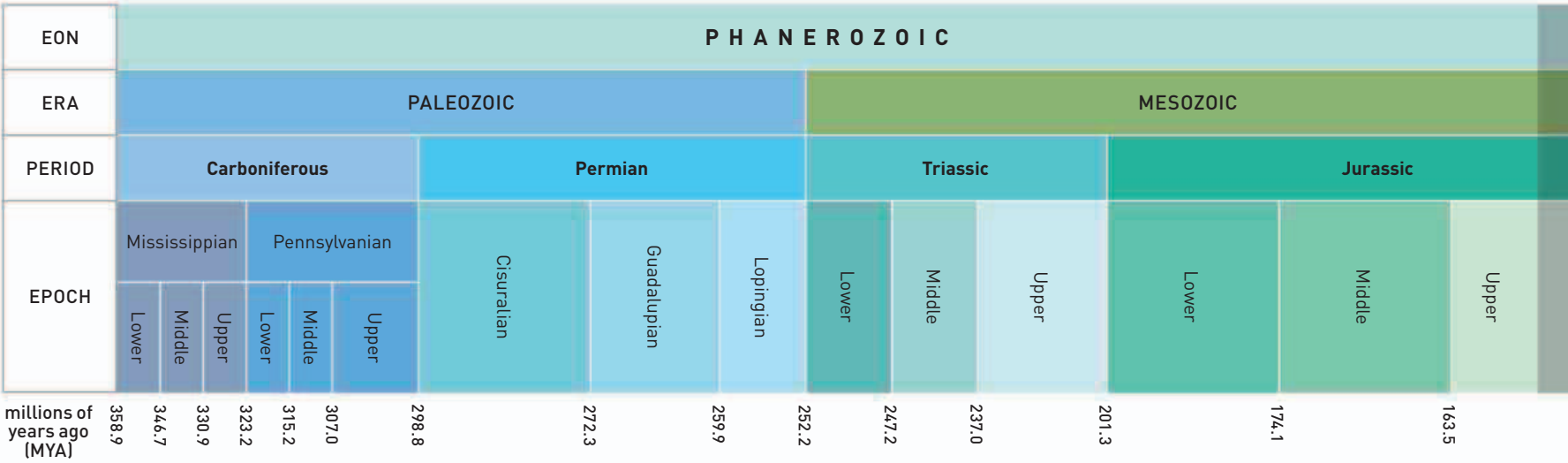
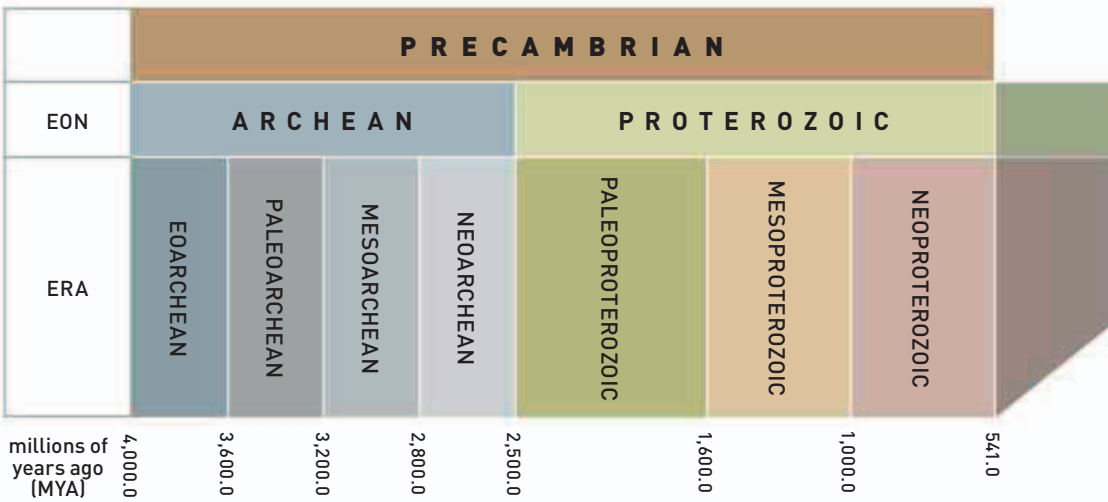


EARTH SCIENCE

THE GEOLOGICAL TIMESCALE

This timescale provides scientists with an internationally recognized chronology of Earth’s history over 4 billion years. The history of Earth is divided into a hierarchical system of named units: the largest are called eons, followed in order of size by eras, periods, epochs, and ages (the latter are not included on the chart below). The timescale allows geologists to go anywhere in the world, examine the rock strata, identify the fossils within them, and give them an approximate age as they know that they are all referring to the same events, strata, and time periods.

The timescale has been developed by examining the history of global changes in ocean and atmospheric chemistry preserved in sedimentary rocks, as well as several other lines of evidence. Lithostratigraphy looks at sedimentary rock types and sequences. Biostratigraphy examines fossils—fossils in the same layer can be matched up across the world. Chronostratigraphy, or radiometric dating, calculates when certain minerals were crystalized, while magnetostratigraphy is a tool that uses the record of the changing polarity of Earth’s magnetic field.



MINERAL CLASSIFICATION

Most minerals are solid, naturally occurring inorganic materials with well-defined chemical compositions and characteristic crystal structures. More than 4,000 are known, although only about 100 are abundant. Minerals are classified according to their chemical composition, and are commonly divided into the groups listed below.

GROUP	APPROXIMATE MINERALS	EXAMPLES
Sulfides	600	Pyrite, galena
Silicates	500	Olivine, quartz, feldspar, garnet
Oxides and hydroxides	400	Chromite, hematite
Phosphates and vanadates	400	Apatite, carnotite
Sulfates	300	Anhydrite, barite, gypsum
Carbonates	200	Calcite, aragonite, dolomite
Halides	140	Fluorite, halite, sylvite
Borates and nitrates	125	Borax, colemanite, kernite, nitratine
Molybdates and tungstates	42	Wulfenite, wolframite
Native elements	20	Gold, platinum, copper, sulfur, carbon

EARTH'S ROCK TYPES

Rocks are naturally occurring assemblages of minerals. All of Earth’s rocks can be categorized as one of the three main types: igneous, sedimentary, and metamorphic. Within each type, geologists recognize many different rocks. Much of this rocky materials is also recycled over geological time.

TYPE	DESCRIPTION
Igneous	Rocks formed by cooling and crystallizing lava or magma. They range from quick-cooled, fined-grained volcanic lava to coarse-grained rocks that have cooled more slowly.
Sedimentary	Rocks formed by the deposition of material on Earth’s surface. Weathering and erosion of rock transports sediment to inland areas, where it is laid down in layers. Plant and animal fossils are found in sedimentary rock.
Metamorphic	When igneous or sedimentary rock is subjected to high temperature and pressure, it is pushed into Earth’s crust, which causes it to flow and recrystallize as metamorphic rock.